# Bayes' Theorem

COMP2711 Discrete Mathematical Tools

Mingxun Zhou (mingxunz@ust.hk)

# Review: Outcomes, events, and probability

| | |
|---|---|
| **Outcome** $\omega$ | One possible result of an experiment. |
| **Sample space** $S$ | The set of all possible outcomes. |
| **Event** $E \subseteq S$ | A collection of outcomes; $E$ happens when $\omega \in E$. |

# Review: Outcomes, events, and probability

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**Classical model:** finite, nonempty $S$; equally likely outcomes.

$$\Pr[E] = \frac{|E|}{|S|}.$$

**General discrete model:** add the probabilities in $E$.

$$\Pr[E] = \sum_{\omega \in E} \Pr[\{\omega\}].$$

# Review: Probability rules

**Conditional probability** $\Pr[E \mid F] = \dfrac{\Pr[E \text{ and } F]}{\Pr[F]}$

**Multiplication rule** $\Pr[E \text{ and } F] = \Pr[F] \Pr[E \mid F]$

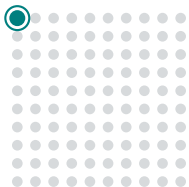
**Independence (definition)** $E, F$ are independent $\iff \Pr[E \text{ and } F] = \Pr[E] \Pr[F]$

**Total probability** $\Pr[E] = \sum_i \Pr[E \mid F_i] \Pr[F_i]$

The $F_i$ are disjoint cases covering the sample space.

Conditioning events must have positive probability.

# Recall: A medical test



**1 in 100** has disease.

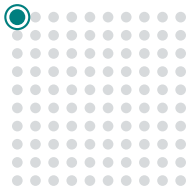
$D$: has the disease. $T$: tests positive.

| | |
|---|---|
| Has the disease | $\Pr[D] = 1\%$ |
| Positive, given disease | $\Pr[T \mid D] = 95\%$ |
| Positive, given no disease | $\Pr[T \mid \overline{D}] = 10\%$ |

**Total probability**

$$\begin{aligned}\Pr[T] &= \Pr[D]\Pr[T \mid D] + \Pr[\overline{D}]\Pr[T \mid \overline{D}] \\ &= 0.01 \cdot 0.95 + 0.99 \cdot 0.10 = \mathbf{10.85\%}.\end{aligned}$$

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**Given a positive test, how likely is the person to have the disease?**

$\Pr[D \mid T] = ?$

# Prior and posterior

## Prior

Before using the test result:

$$\Pr[D] = 1\%.$$

Evidence: the test is positive

# Prior and posterior

**Prior**

Before using the test result:

$$\Pr[D] = 1\%.$$

**Posterior**

After observing a positive result:

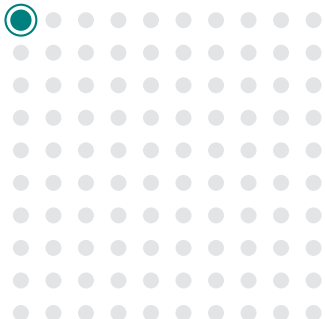
$$\Pr[D \mid T] = ?$$

Evidence: the test is positive

Our information changes, not the person's disease status.

# Among the positive results

$\Pr[D] = 1\%, \quad \Pr[T \mid D] = 95\%, \quad \Pr[T \mid \overline{D}] = 10\%.$



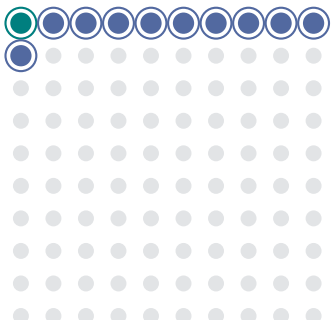
Approximate picture of 100 people.

**Positive with disease:**

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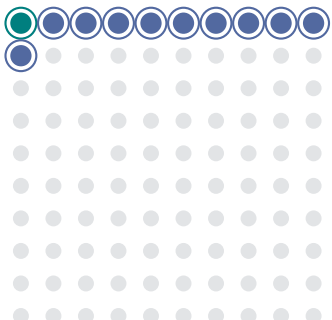
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# Among the positive results

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Approximate picture of 100 people.

**Positive with disease:**

$$100(0.01)(0.95) = 0.95 \approx 1.$$

**Positive without disease:**

$$100(0.99)(0.10) = 9.9 \approx 10.$$

**Only a small fraction of positives have the disease.**

$$\Pr[D \mid T] = \frac{0.95}{0.95 + 9.9} \approx 8.76\%.$$

# A positive result changes the probability

**Prior**

$\Pr[D] = 1\%$

Positive result →

**Posterior**

$\Pr[D \mid T] \approx 8.76\%$

**Surprisingly:** The test is over 90% correct overall in this population, yet after a positive result, the chance of disease is **less than 10%!**

# A positive result changes the probability

**Prior** $\Pr[D] = 1\%$ —Positive result→ **Posterior** $\Pr[D \mid T] \approx 8.76\%$

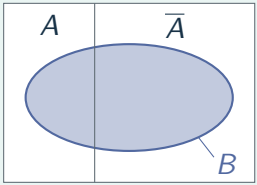
**Surprisingly:** The test is over 90% correct overall in this population, yet after a positive result, the chance of disease is **less than 10%!**

$$\underbrace{\Pr[T \mid D]}_{\text{positive, given disease}} = 95\% \quad \neq \quad \underbrace{\Pr[D \mid T]}_{\text{disease, given positive}} \approx 8.76\%.$$

# Bayes' theorem

**MUST UNDERSTAND**

For $0 < \Pr[A] < 1$ and $\Pr[B] > 0$:



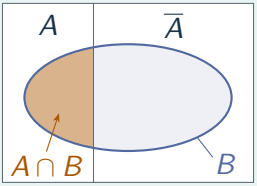

By conditional probability definition:

$$\Pr[A \mid B] = \frac{\Pr[A \text{ and } B]}{\Pr[B]}$$

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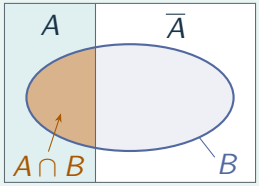

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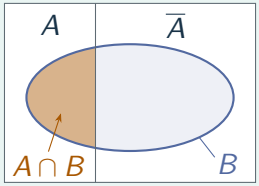
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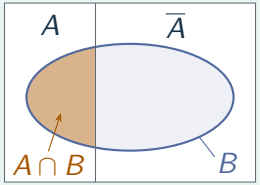
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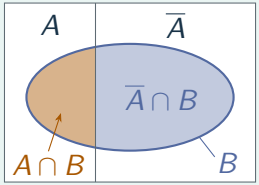
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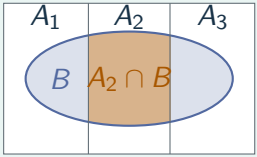
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# Bayes' theorem: Multiple disjoint cases

Let $A_1, \ldots, A_n$ be **disjoint cases covering the sample space**.

Assume $\Pr[A_j] > 0$ for every $j$, and $\Pr[B] > 0$.



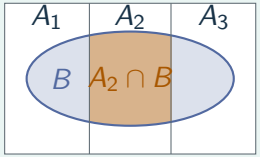

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$$\Pr[B] = \sum_{j=1}^{n} \Pr[B \mid A_j] \Pr[A_j].$$

$$\Pr[A_i \mid B] = \frac{\Pr[B \mid A_i] \Pr[A_i]}{\sum_{j=1}^{n} \Pr[B \mid A_j] \Pr[A_j]}.$$

Among all cases producing evidence $B$, what fraction comes from $A_i$?

# When to use Bayes' theorem: Inference

**Bayes' theorem is a tool for inference:**
**using observed evidence to assess possible explanations.**

### What we know

If $A$ is true, how likely is evidence $B$?

$$\Pr[B \mid A]$$

# When to use Bayes' theorem: Inference

**Bayes' theorem is a tool for inference:**
**using observed evidence to assess possible explanations.**

**What we know**

If $A$ is true, how likely is evidence $B$?

$$\Pr[B \mid A]$$

**What we want to infer**

We observe $B$. How likely is $A$?

$$\Pr[A \mid B]$$

**Alarm:** We hear an alarm. **How likely is there a fire?**

**Email:** An email says "1 million dollar prize." **How likely is it spam?**

# Example: Three factories

A company receives products from three factories.

$F_A, F_B, F_C$: the product comes from factory $A, B, C$, respectively.

| Factory | Share of all products | Defect rate |
|---|---|---|
| $A$ | 50% | 2% |
| $B$ | 25% | 4% |
| $C$ | 25% | 8% |

Select one product uniformly at random. Let $D$ be the event it is defective.

**Given $D$, what is the probability it came from factory $C$?**

That is, find $\Pr[F_C \mid D]$.

# Three factories: Calculation

$D$: defective. $F_A, F_B, F_C$: comes from factory $A, B, C$.

| Factory | Prior | Defect rate |
|---|---|---|
| $A$ | 0.50 | 0.02 |
| $B$ | 0.25 | 0.04 |
| $C$ | 0.25 | 0.08 |

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| Factory | Prior | Defect rate | Defective and from factory |
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| $B$ | 0.25 | 0.04 | $0.25 \cdot 0.04 = 0.01$ |
| $C$ | 0.25 | 0.08 | $0.25 \cdot 0.08 = 0.02$ |

$$\Pr[D] = \sum_{i \in \{A,B,C\}} \Pr[D \mid F_i] \Pr[F_i] = 0.01 + 0.01 + 0.02 = 0.04.$$

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$$\Pr[F_C \mid D] = \frac{\Pr[D \mid F_C] \Pr[F_C]}{\Pr[D]} = \frac{0.02}{0.04} = \frac{1}{2} = 50\%.$$

# Bayes' theorem: Summary

**Prior** $\Pr[A]$ — Observe $B$ → **Posterior** $\Pr[A \mid B]$

$$\Pr[A \mid B] = \frac{\Pr[B \mid A] \Pr[A]}{\Pr[B]}.$$

Use total probability to find $\Pr[B]$ when needed.